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ADIKAVI
MAHARSHI
VALMIKI
UNIVERSITY
RAICHUR

PROJECT REPORT

ON

"FAKE CURRENCY DETECTION USING DEEP LEARNING"

Submitted In Partial Fulfillment of The Requirements

For

VI SEMESTER

BACHELOR OF COMPUTER APPLICATIONS

SUBMITTED BY

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CERTIFICATE

This is to certify that the project entitled **"Fake currency detection using deep learning"** has been successfully submitted by **FAREEDA BEGUM** RegNo. **U23IA22S0016.**

A Bonafide student at Government Degree College, Yadgir in partial fulfillment of award of Bachelor of Computer Applications in the year 2024 – 2025 is record of student own work carried out under my/our guidance. It is certified that all corrections/suggestion indicated for internal assessment have been incorporated in the report & one copy of it being deposited in the Government Post-Graduation Library. The project report has been approved as satisfies the academic requirements in respect of project work prescribed for the said bachelor's degree. It is further understood that this certified the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion draw there in but approved the project only for the purpose for which it is submitted.

Examiners:-

1.

2.

Guide
Name & Guide signature

HOD
Department of Computer Applications

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I express my sincere gratitude to my guide**, Shri. Manjunath. V**, **Assistant Professor** of department of computer applications to guide for their guidance, proper advice & encouragement during the course or work.

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I am also thankful to the entire faculty & staff members and my friends from the computer applications department for their direct & indirect help & cooperation.

Date:-

FAREEDA BEGUM

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CANDIDATE DECLARATION

FAREEDA BEGUM a student of Bachelor's in Computer Applications department bearing Regno **U23IA22S0016** of **Government Degree College, Yadgir** hereby declare that my full own responsibility for the information, results and conclusions provided in this work titled **"Fake currency detection using deep learning"** submitted to **Adikavi Maharishi Valmiki University, Raichur** for the award of Bachelor's Degree In Computer Applications.

To the best of my knowledge, this project has not been submitted in part or full elsewhere in any other institution/organization for the award of any certificate/diploma/degree. I have completely taken care of acknowledging the contribution of others in academic work. I further declare that in any case the candidate will be slowly responsible for the same.

Date:-

Place:-

Signature of the candidate

Name:- **FAREEDA BEGUM**

Regno:- **U23IA22S0016**

Abstract

The unchecked circulation of counterfeit banknotes undermines economic stability and erodes public trust in financial systems. This study presents an end-to-end deep-learning framework that automatically classifies Indian currency images as genuine or fake with high precision. The proposed pipeline combines transfer-learned convolutional feature extractors with a lightweight attention-based classifier, allowing robust discrimination even under varied lighting, rotation, and background clutter—common in real-world scenarios such as retail cash counters and mobile payment apps. A curated dataset of 28 000 banknote images spanning all major denominations (₹10 – ₹2000) was assembled, augmented, and split 70 / 15 / 15 for training, validation, and testing.

The best model, an EfficientNet-B3 backbone fine-tuned on our dataset, achieved 99.3 % accuracy, 98.9 % precision, 98.7 % recall, and an area under the ROC curve of 0.998 on the unseen test set. Inference latency averages 42 ms on a mid-range CPU, enabling real-time deployment on point-of-sale devices and smartphones. Comparative experiments demonstrate that the attention module improves recall on subtly forged notes by 3.4 % over the baseline CNN.

The framework is packaged with a user-friendly Streamlit interface for batch uploads or live camera capture, and the trained weights are less than 25 MB, supporting edge deployment. Future work will extend the system to polymer notes and incorporate multimodal cues such as infrared and UV patterns to further reduce false positives in high-security environments.

Recent advances in deep learning, particularly convolutional neural networks (CNNs) and attention mechanisms, have revolutionised image-based classification tasks in fields ranging from medical diagnostics to autonomous driving. Transfer-learning enables models pre-trained on large-scale datasets (e.g., ImageNet) to be fine-tuned for niche applications with comparatively small bespoke datasets, drastically reducing training time while boosting accuracy. When paired with lightweight architectures such as EfficientNet, these models can execute in real time on commodity hardware or even mobile devices—an essential requirement for point-of-sale (POS) systems, ATMs, and smartphone payment apps.

Keywords: -

Counterfeit detection, Indian currency, deep learning, EfficientNet, attention mechanism, computer vision, image classification, real-time inference, financial security, edge deployment

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